Sunggyu Ryoo

Ph.D. Student, Department of Physics and Astronomy, Seoul National University

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EDUCATION

Mar 2024–Present



Ph.D. Seoul National University
Department of Physics and Astronomy
Supervised by Prof. Takhee Lee

Mar 2020–Feb 2024



B.S. Seoul National University
Department of Physics and Astronomy
Summa Cum Laude

SCHOLARSHIPS

2024–Present

SBS Foundation Scholarship
SBS, The top merit-based private scholarship for graduate students

2020–2023

Korean Presidential Scholarship for Science
Korea Student Aid Foundation, The top merit-based scholarship for undergraduate students

RESEARCH INTERESTS

- 2D Electronics & Optoelectronics
- Device Physics & Engineering
- Noise & Transport Physics

AWARDS AND ACHIEVEMENTS

2025–2026

Outstanding Research Award
Department of Physics and Astronomy, Seoul National University
Academic achievements in Ph.D. research (Spring 2025, Spring 2026)

2024

University President's Award (Excellence Award)
Faculty of Liberal Education, Seoul National University
Academic achievements in undergraduate research

2023

The Grand Award
Department of Physics and Astronomy, Seoul National University
Academic achievements in undergraduate research

2020–2021

Dean's List
College of Natural Sciences, Seoul National University
Academic achievements in undergraduate exams (Spring 2020, Fall 2020, Spring 2021, Fall 2021)

PUBLICATIONS

† Equal contribution · * Corresponding author

01 **Noise-Suppressed High-Performance WSe_2 Phototransistors enabled by Optically Programmed Remote Contact Doping**

Sunggyu Ryoo, Seongmin Ko, Juntae Jang, Donguk Kim, Jaehyoung Park, Yeeun Kim, Youngmin Song, Jaeyong Woo, Jongeun Yoo, Kyungjune Cho*, and Takhee Lee*

ACS Nano, 20, 29045 (2026).

02 **Remote Charge Modulation in WSe_2 Transistors via Plasma-Induced Charged Defects in hBN**

Seongmin Ko†, **Sunggyu Ryoo**†, Juntae Jang, Jaeyong Woo, Donguk Kim, Jaehyoung Park, Jongeun Yoo, Young Duck Kim*, Kyungjune Cho*, and Takhee Lee*

ACS Applied Electronic Materials, 8, 9554 (2026).

03 **Impact of Gate-Induced Mobility Attenuation on Accurate Characterization of Perovskite Field Effect Transistors**

Youngmin Song, Jaeyong Woo, Hyemin Lee, Jaehyoung Park, Seongmin Ko, Yeeun Kim, **Sunggyu Ryoo**, Henning Sirringhaus, Keehoon Kang*, and Takhee Lee*

ACS Applied Materials & Interfaces, 18, 58058 (2026).

04 **Dopant- and Trap-Limited Gate Modulation in 2D Tin Halide Perovskite Field-Effect Transistors**

Jaeyong Woo, Seongmin Ko, Youngmin Song, Yeeun Kim, **Sunggyu Ryoo**, Taehyun Kong, Keehoon Kang*, and Takhee Lee*

ACS Applied Electronic Materials, 8, 6156 (2026).

05 **Photoresponse Properties of Ambipolar Transport in WSe_2 Field-Effect Transistors**

Jongeun Yoo†, Seongmin Ko†, **Sunggyu Ryoo**†, Jaehyoung Park, Jaeyoung Kim, Juntae Jang, Jinwoo Sim, Donguk Kim, SeungNam Cha, Chongwu Zhou, Kyungjune Cho*, and Takhee Lee*

Advanced Electronic Materials, 12, e00669 (2026).

06 **Strong-Binding Small-Molecule Passivator for Two-Dimensional Tin-Based Perovskite Field-Effect Transistors**

Hyeonmin Choi†, Joonha Jung†, Young-Kwang Jung†, Seok Woo Lee, Taehyun Kong, Yongjin Kim, Youjin Reo, Jinwoo Sim, Yeeun Kim, Jaeyong Woo, **Sunggyu Ryoo**, Jaeyoon Cho, Youcheng Zhang, Stefano Pecorario, Bo Ram Lee, Henning Sirringhaus, Yong-Young Noh, Samuel Stranks, Takhee Lee*, and Keehoon Kang*

Advanced Functional Materials, 36, e74094 (2026).

07 **Enhancing Photodetection of WSe_2 Phototransistors by Reducing Charge Scattering with hBN Passivation**

Sunggyu Ryoo†, Taehee Kim†, Jongeun Yoo, Seongmin Ko, Kyungjune Cho*, and Takhee Lee*

Nano Futures, 9, 035001 (2025).

08 **Noise-Reduced WSe_2 Phototransistors for Enhanced Photodetection Performance via Suppression of Metal-Induced Gap States**

Sunggyu Ryoo, Jinwoo Sim, Seungjun Jeong, Juntae Jang, Jaeyong Woo, Jaehyoung Park, Seongmin Ko, Yeeun Kim, Youngmin Song, Jongeun Yoo, Heebeom Ahn, Keehoon Kang, Daeheum Cho, Kyungjune Cho*, and Takhee Lee*

Advanced Materials Technologies, 10, 2500064 (2025).

PUBLICATIONS

† Equal contribution · * Corresponding author

09 **Enhanced Photodetection Performance of In Situ Core/Shell Perovskite-MoS_2 Phototransistor**

Jinwoo Sim†, **Sunggyu Ryoo**†, Joo Sung Kim†, Juntae Jang, Heebeom Ahn, Donguk Kim, Joonha Jung, Taehyun Kong, Hyeonmin Choi, Yun Seog Lee, Tae-Woo Lee*, Kyungjune Cho*, Keehoon Kang*, and Takhee Lee*

ACS Nano, 18, 16905 (2024).

PRESENTATIONS

01 **Optical Contact Doping for Low-Noise WSe_2 Phototransistors**

Sunggyu Ryoo, Seongmin Ko, Juntae Jang, Donguk Kim, Jaehyoung Park, Youngmin Song, Yeeun Kim, Jaeyong Woo, Jongeun Yoo, Kyungjune Cho, and Takhee Lee

The 34th Korean Conference on Semiconductors (KCS 2027), Gangneung Convention Center, Gangneung, Korea, January 26-29, 2027.

Oral presentation

02 **Photo-Induced Contact Doping for Low-Noise and High-Performance WSe_2 Phototransistors**

Sunggyu Ryoo, Seongmin Ko, Juntae Jang, Donguk Kim, Jaehyoung Park, Youngmin Song, Yeeun Kim, Jaeyong Woo, Jongeun Yoo, Kyungjune Cho, and Takhee Lee

2026 MRS (Materials Research Society) Fall Meeting, Hynes Convention Center, Boston, USA, November 29-December 4, 2026.

Oral presentation

03 **High-Performance WSe_2 Phototransistors with Optically Programmed Remote Contact Doping**

Sunggyu Ryoo, Seongmin Ko, Juntae Jang, Donguk Kim, Jaehyoung Park, Youngmin Song, Yeeun Kim, Jaeyong Woo, Jongeun Yoo, Kyungjune Cho, and Takhee Lee

2026 KPS (Korean Physical Society) Fall Meeting, Jeju International Convention Center, Jeju, Korea, October 20-23, 2026.

Oral presentation

04 **Noise Reduction in WSe_2 Phototransistors for Enhanced Photodetection**

Sunggyu Ryoo, Jinwoo Sim, Taehee Kim, Juntae Jang, Seungjun Jeong, Daeheum Cho, Kyungjune Cho, and Takhee Lee

The 33rd Korean Conference on Semiconductors (KCS 2026), High One resort, Jeongseon, Korea, January 27-30, 2026.

05 **High-Performance Phototransistor based on In-Situ Core/Shell Perovskite-MoS_2 Heterostructure**

Sunggyu Ryoo, Jinwoo Sim, Joo Sung Kim, Tae-Woo Lee, Kyungjune Cho, Keehoon Kang, and Takhee Lee

The 15th International Conference on Flexible and Printed Electronics (ICFPE 2025), Institute of Science Tokyo, Japan, September 17-19, 2025.

Best Poster Award

06 **High-Performance MoS_2 Phototransistor using In-Situ Core/Shell Perovskite with Photogating Effect**

Sunggyu Ryoo, Jinwoo Sim, Joo Sung Kim, Tae-Woo Lee, Kyungjune Cho, Keehoon Kang, and Takhee Lee

International Symposium on Organic and Perovskite Electronics (ISOPE 2025), Seoul National University, Seoul, Korea, July 30-August 1, 2025.

PRESENTATIONS

07 **Performance-Enhanced WSe_2 Phototransistors through Noise Reduction using Semimetal Contacts**

Sunggyu Ryoo, Jinwoo Sim, Seungjun Jeong, Juntae Jang, Keehoon Kang, Daeheum Cho, Kyungjune Cho, and Takhee Lee

2025 KPS (Korean Physical Society) Spring Meeting, Daejeon Convention Center, Daejeon, Korea, April 23-25, 2025.

Oral presentation

08 **High-Performance Photodetection of In-Situ Core/Shell Perovskite-MoS_2 Phototransistors**

Sunggyu Ryoo, Jinwoo Sim, Joo Sung Kim, Juntae Jang, Heebeom Ahn, Tae-Woo Lee, Keehoon Kang, Kyungjune Cho, and Takhee Lee

APS (American Physical Society) Global Physics Summit 2025 March Meeting, Anaheim Convention Center, Anaheim, USA, March 16-21, 2025.

Oral presentation

09 **Semimetal Contact Engineering for Low-Noise, High-Performance WSe_2-Based Phototransistors**

Sunggyu Ryoo, Jinwoo Sim, Jaeyong Woo, Jaehyoung Park, Yeeun Kim, Youngmin Song, Heebeom Ahn, Kyungjune Cho, and Takhee Lee

The 32nd Korean Conference on Semiconductors (KCS 2025), High One resort, Jeongseon, Korea, February 12-14, 2025.

Oral presentation

10 **Design and Fabrication of an In-situ Core/Shell Perovskite/MoS_2 Heterostructure for High-Performance Photodetection**

Sunggyu Ryoo, Jinwoo Sim, Joo Sung Kim, Juntae Jang, Tae-woo Lee, and Takhee Lee

The 31st Korean Conference on Semiconductors (KCS 2024), Gyeongju Hwabaek International Convention Center, Gyeongju, Korea, January 24-26, 2024.

Oral presentation

11 **Enhancing Photodetection Performance with In-Situ Core/Shell Perovskite-MoS_2 Heterostructure**

Sunggyu Ryoo, Jinwoo Sim, Joo Sung Kim, Juntae Jang, Tae-Woo Lee, and Takhee Lee

NANO KOREA 2023, KINTEX, Ilsan, Korea, July 5-7, 2023.

TEACHING

Fall 2025	**Teaching Assistant** (College Physics 2) Seoul National University, **Excellent TA Award**
Spring 2025	**Teaching Assistant** (College Physics 1) Seoul National University
Fall 2024	**Teaching Assistant** (Physics Lab 2) Seoul National University, **Excellent TA Award**
Spring 2024	**Teaching Assistant** (Physics Lab 1) Seoul National University
Fall 2021	**Tutor** (College Writing 2: Writing in Science & Technology) Seoul National University